Mubashara Rehman

AI Researcher • Computer Vision & Medical Imaging

PhD Thesis Submitted – University of Udine, Italy | IRCCS Centro di Riferimento Oncologico di Aviano, Italy

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Website | **LinkedIn** | **Google Scholar** | **ORCID**

Professional Profile

PhD researcher in Artificial Intelligence with expertise in computer vision, deep learning and medical image analysis. Experienced in designing and developing generative AI models for CT and MRI reconstruction, image enhancement and cross-modality domain translation using PyTorch. Proven ability to collaborate across academia, healthcare and industry (including IRCCS CRO Aviano (Italy), Leiden University Medical Center (LUMC), and Philips Research) to deliver innovative AI solutions for medical imaging. Looking ahead, I aim to further advance my research in AI while exploring emerging areas such as Neuro-AI, multimodal learning, foundation models, and intelligent systems, with a focus on developing AI that can address complex real-world challenges.

Research Interests

- Deep Learning for Medical Imaging
- Metal Artifact Reduction in CT
- CT Domain Transformation (kVCT → MVCT)
- AI-driven CT Reconstruction & MRI Enhancement

Education

Ph.D. in Artificial Intelligence and Data Analysis for Personalised Oncology — Sep 2023 – Aug 2026

Università degli Studi di Udine | Udine, Italy

Thesis submitted August 2026 • Dept. of Mathematics, Computer Science and Physics (DMIF), in collaboration with IRCCS CRO Aviano

Advisor: Prof. Christian Micheloni • Co-Advisor: Prof. Niki Martinel (Machine Learning and Perception Lab)

Collaborators: Dr. Michele Avanzo, Dr. Riccardo Spizzo (IRCCS CRO Aviano)

- Research on deep learning methods for metal artifact reduction in CT images for radiotherapy applications.
- Developed kVCT-to-MVCT domain transformation frameworks to improve image quality and support radiotherapy treatment planning.

M.Sc. in Software Engineering — Sep 2017 – Nov 2021

Tianjin University, College of Intelligence and Computing | Tianjin, China

- Graduated with 83.2%; awarded the Tianjin University International Student Scholarship.

B.Sc. in Software Engineering — Sep 2012 – Jun 2016

Fatima Jinnah Women University | Rawalpindi, Pakistan

- Final grade: 82%; degree verified by World Education Services (WES).
- Senior design thesis: *"Design and Development of an Android Application for Indoor Navigation and Positioning System using Sensor Networks"* – graded A+.

Higher Secondary School Certificate (HSSC) – Pre-Engineering — 2010 – 2012

Grades 11 and 12 | Grade A

Secondary School Certificate (SSC) – Science — 2008 – 2010

Grades 9 and 10 | Grade A

Research Experience

PhD Visiting Researcher — Jan 2026 – Jun 2026

Leiden University Medical Center (LUMC) | Leiden, Netherlands

Supervisor: Prof. Marius Staring • In collaboration with Dr. Mariya Doneva (Philips Research) and Dr. Yiming Dong (LUMC)

Partners: Division of Image Processing (LKEB) and C.J. Gorter Center for High Field MRI, LUMC; Philips Research, Netherlands

- Researched deep learning-based low-field to high-field prostate MRI enhancement in a joint LUMC–Philips Research project.

- Developed novel generative AI methods for MRI reconstruction and enhancement, focused on preserving anatomical structures and improving image quality.
- **Outcome:** First author of *"ReFineMRI"*, submitted to *IEEE Transactions on Computational Imaging* (2026).

Research Collaboration Apr 2024 – Sep 2024

Universidad de Santiago de Compostela | Santiago de Compostela, Spain

With Dr. Belén Serrano-Antón and Prof. Alberto P. Muñuzuri • Nonlinear Physics Group, CITMAga, FlowReserve Labs S.L.

- Worked on medical imaging research focused on metal artifact reduction and CT domain transformation.
- **Outcome:** First author of *"MAR-DTN"*, published at the 27th International Conference on Pattern Recognition (ICPR 2024).

Publications

Journal Articles

[1] M. Avanzo, A. Corso, **M. Rehman**, C. Micheloni, N. Martinel, R. Spizzo, C. Deana, G. Fanetti, C. Cappelletto, P. Chiovati, G. Pirrone, S. Peric, A. Dassie, "Evaluation of Quality of Synthetic Megavolt Computed Tomography (MVCT) for Planning of Radiotherapy Treatments of Head and Neck," *Physica Medica: European Journal of Medical Physics*, vol. 142, Feb. 2026.

[2] D. Spina, K. Roitero, S. Mizzaro, V. Della Mea, F. Da Ros, M. Soprano, **M. Rehman**, et al., "Report on the Hands-On PhD Course on Responsible AI from the Lens of an Information Access Researcher," *ACM SIGIR Forum*, vol. 58, no. 2, pp. 1–61, Mar. 2025.

Conference Papers

[3] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "H3D-MarNet: Wavelet-Guided Dual-Path Learning for Metal Artifact Suppression and CT Modality Transformation for Radiotherapy Workflows," in *Pattern Recognition – ICPR 2026*, LNCS vol. 16821, Springer, pp. 123–138, 2026.

[4] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "ReMAR-DS: Recalibrated Feature Learning for Metal Artifact Reduction and CT Domain Transformation," *23rd International Conference on Image Analysis and Processing (ICIAP)*, Sep. 2025.

[5] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "MAR-DTN: Metal Artifact Reduction using Domain Transformation Network for Radiotherapy Planning," *27th International Conference on Pattern Recognition (ICPR)*, 2024.

Under Review

[6] **M. Rehman**, D. Lyu, M. Staring, N. Martinel, M. Doneva, C. Micheloni, Y. Dong, "ReFineMRI: Structure-Preserving Low-Field MRI Enhancement via Progressive Refinement," submitted to *IEEE Transactions on Computational Imaging (TCI)*, 2026.

[7] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "WAVEFORM-Net: Wavelet-Enhanced Frequency Fusion for Transformer-Based Metal Artifact Reduction in CT Domain Translation," submitted to *Multimedia Tools and Applications* (Springer), 2026.

Technical Skills

- **Programming:** Python, C/C++
- **AI & Computer Vision:** PyTorch, OpenCV, NumPy, Pandas, scikit-image, Matplotlib, Weights & Biases (W&B)
- **Core Expertise:** Deep Learning, Computer Vision, Medical Image Analysis, Generative AI, CNNs, Vision Transformers (ViTs), Diffusion Models, Flow Matching, Image Reconstruction, Image Enhancement, Cross-Modality Domain Translation
- **Medical Imaging:** CT, MRI, DICOM, 3D Slicer (advanced), ITK-SNAP (basic)
- **Tools & Platforms:** Git, PyCharm, Jupyter Notebook / JupyterLab, Draw.io, Jira, Miro
- **Research & Documentation:** LaTeX, BibTeX, EndNote, Mendeley

Professional Experience

Research Collaborator Sep 2023 – Sep 2026

IRCCS Centro di Riferimento Oncologico di Aviano | Aviano, Italy

- Collaborated with oncologists and medical physicists on CT imaging datasets.

- Developed AI-driven methods for Metal Artifact Reduction (MAR) and cross-modality domain transformation (kVCT → MVCT).
- Integrated deep learning models for CT reconstruction and enhancement into radiotherapy workflows.

IT Project Lead and CRM Manager May 2021 – Jun 2023
Moreno Holdings Pvt Ltd. | Islamabad, Pakistan

Senior Research Executive Oct 2016 – Dec 2019
Empower Pakistan | Pakistan (Remote, Part-time)

Intern Oct 2015 – May 2016
Crevo Technologies | Islamabad, Pakistan

Intern Aug 2015 – Sep 2015
Ministry of Defence Production, Directorate of Procurement (Navy) | Rawalpindi, Pakistan

Awards and Honours

- **PhD Fellowship** – Fully funded doctoral fellowship from the Italian Ministry of Health, in collaboration with the National Cancer Research Institute, CRO IRCCS. 2023
- **Master's Scholarship** – Tianjin University International Student Scholarship, for academic excellence. 2017
- **Laptop Award** – Prime Minister's Best Student Laptop Scheme, Government of Pakistan. 2015

Summer Schools and Research Training

- **RISE-MICCAI Summer School 2025** – MICCAI Society Jul 2025
- **International Summer School on AI (AI-DLDA), 8th Edition** – University of Udine Jun 2025
- **International Computer Vision Summer School (ICVSS 2024)** – University of Cambridge Aug 2024
- **International Summer School on AI (AI-DLDA), 7th Edition** – University of Udine Jul 2024
- **Summer School for Humans and Technology: Technologies for Good** – DAAD, Germany Sep 2023

Professional Affiliations

- **Member**, Italian Association for Research in Computer Vision, Pattern Recognition and Machine Learning (CVPL), affiliated with the IAPR. 2025 – Present

Languages

- **English** – Fluent
- **Urdu** – Native

Volunteering and Activities

- **Soprano Singer** (Nov 2019) – International Students Chorus, Tianjin University; performed to represent cultural exchange through music.
- **Cultural Volunteer** (Oct 2018) – Chinese Opera Culture Week (Peking Opera), 9th China (Beijing) International Garden Expo.
- **Volunteer** – Esmile Khursheed Welfare, Rawalpindi; ran extra-curricular and confidence-building sessions with students.
- **Volunteer** – Al-Shifa Eye Trust Hospital, Rawalpindi; assisted staff with patient handling during daily operations.

Hobbies & Interests

- **Travel & Exploration:** Discovering new places, experiencing different cultures, collecting souvenirs and exploring local cuisines.
- **Fitness & Outdoors:** Fitness, athletics, gardening and fishing.

References

Available upon request.